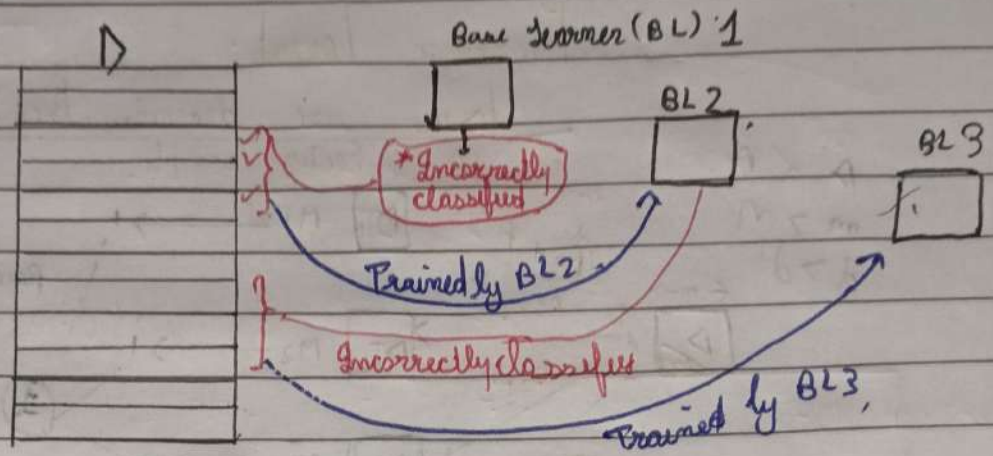


06-09-2025 !!

What is Boosting (ADA Boost)

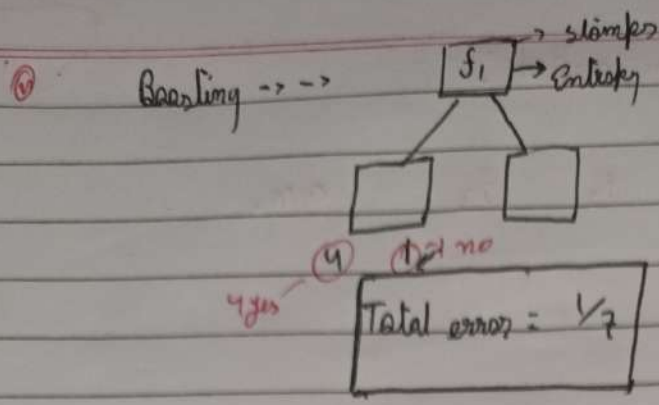


In ADA Boost. $\rightarrow$ we use weights i.e assigned for proper training - classification.

$n = \text{no. of records}$

$w = 1/n$

	f_1	f_2	f_3	O/P.	Sample weight,
1	_____	_____	_____		$1/7$
2	_____	_____	_____		$1/7$
3	_____	_____	_____		$1/7$
4	_____	_____	_____		$1/7$
5	_____	_____	_____		$1/7$



iii) Performance of stamp = $\frac{1}{2} \log_e \left(\frac{1 - \text{Total error}}{\text{Total error}} \right)$

$$\frac{1}{2} \log_e \left(\frac{1 - \frac{1}{7}}{\frac{1}{7}} \right)$$

$$\frac{1}{2} \log_e [6]$$

$$\Rightarrow 0.896$$

Performance

Say for performance $\rightarrow 0.896$, we will create a column which has * Updated weights *. Say, if predⁱ was right ('yes') we decrease weight, if ('No') we increase the weight.

Updated samp		Abn wei	New Sample weight for every
yes	* 1 $\downarrow$ $\rightarrow 0.05$	0.07	$\Rightarrow$ $\frac{1}{7} \times e^{0.895}$ $\Rightarrow 0.349$
no	* 2 $\rightarrow 0.349$	0.513	
yes	* 3 $\uparrow$ * 0.05	0.07	$\left. \begin{array}{l} \text{normalized} \\ \text{values} \end{array} \right\} \Rightarrow 0.68$
no	* 4 $\uparrow$ $\rightarrow 0.349$	0.513	
	* 5 $\rightarrow 0.05$	0.07	$\Rightarrow 0.349$
	* 6 $\rightarrow 0.05$	0.07	
	* 7 $\rightarrow 0.05$	0.07	$\Rightarrow 0.349$
	* 8 $\rightarrow 0.05$	0.07	

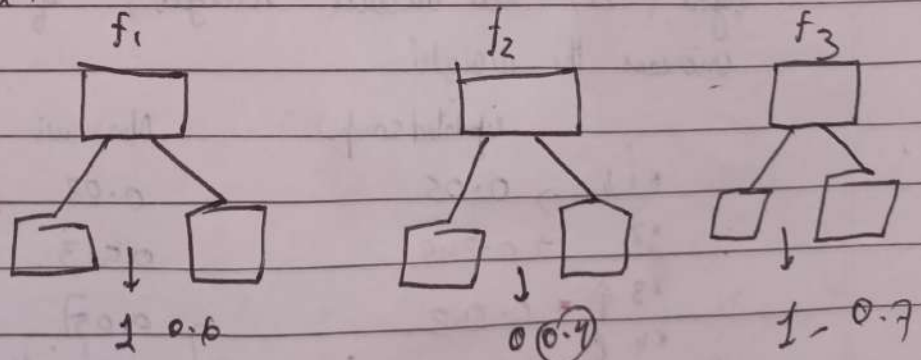
New sample weight for right
 $\Rightarrow$ weight * e $\Rightarrow 0.05$

⑦. new dataset is created:

Normalized weight	f_1	f_2	f_3	o/p
0 to 0.07				
0.07 to 0.58				
0.58 - 0.65				
0.65 - 0.72				
0.72 → 0.79				
0.79 → 0.86				

As to normalized weights, the stumps are created and are made to classify all the irregularities

Result →



∴ Random for Boosting works on majority votes.

So, $R = 1$.